
TDSCSI-2026-04-1278 - Decision (Reject)

Transactions on Dependable and Secure Computing

Fri, May 8, 2026 at

<onbehalf@manuscriptcentral.com>

12:18 AM

Reply-To: rawat.a@ieee.org

To: sinem.sav@cs.bilkent.edu.tr, hkanpak21@ku.edu.tr, akupcu@ku.edu.tr

RE: TDSCSI-2026-04-1278, "HE-IFD: Privacy-Preserving One-Shot Federated Distillation under Homomorphic Encryption"
Manuscript Type: SI-Special Issue on Security and Privacy in Federated Learning and Unlearning

Dear Dr. Sav,

We have completed the review process of the above-referenced paper that was submitted to the IEEE Transactions on Dependable and Secure Computing. Enclosed are your reviews. We hope that you will find the editor's and reviewers' comments and suggestions helpful.

We regret to inform you that, based on the reviewer feedback, the Associate Editor could not recommend publishing your paper. Final decisions are based on the referees' reviews and such factors as restriction of space, topic, and the overall balance of articles.

If you choose to revise and resubmit the manuscript, please note that resubmission should reflect substantial revisions addressing the major concerns raised during the review process and will only be considered after a minimum waiting period of six months.

We hope that this decision does not deter you from submitting to us again. Thank you for your interest in the IEEE Transactions on Dependable and Secure Computing.

Sincerely,

IEEE Transactions on Dependable and Secure Computing

Editor Comments

Associate Editor

Comments to the Author:

Thank you for submitting your manuscript on HE-IFD, which proposes a one-shot federated knowledge distillation framework under homomorphic encryption. The reviewers recognize the relevance of the problem and acknowledge several strengths of the work, including the practical motivation of reducing communication rounds, the clear overall system workflow, and the attempt to combine CKKS-based encrypted training with one-shot federated distillation. Reviewer 1 particularly appreciated the empirical privacy evaluation against membership inference attacks and the communication-computation decoupling property of the framework. However, the reviewers also raised multiple substantial concerns that need to be addressed before the work can be considered for publication. Reviewer 1 noted that **the method suffers from severe performance degradation under larger client counts and stronger non-IID settings, without proposing mechanisms to mitigate this limitation.** The reviewers also emphasized that **the practical communication overhead remains extremely large, and that the manuscript lacks sufficient discussion of deployment feasibility, bandwidth constraints, or compression strategies.** Reviewer 2 identified the absence of direct comparisons with prior federated distillation and encrypted FL methods as a major weakness, making it difficult to assess the true improvement of the proposed method. Important experimental details are also missing, including **end-to-end CKKS training measurements, encryption throughput, memory usage, and convergence behavior.** Reviewer 2 additionally requested stronger analysis regarding malicious or colluding clients, as well as privacy leakage from the released student model. Reviewer 3 pointed out several conceptual and organizational issues, including unclear alignment between the stated challenges and claimed contributions, insufficient explanation of how specific components address the proposed challenges, weak motivation presentation structure, and limited

baseline comparisons. Overall, while the paper presents an interesting direction with potential practical significance, the current manuscript still requires substantial improvements in experimental rigor, technical justification, scalability analysis, and presentation clarity.

Reviewer Comments

Please note that some reviewers may have included additional comments in a separate file. If a review contains the note "see the attached file" under Section III A - Public Comments, you will need to log on to ScholarOne Manuscripts to view the file. After logging in, select the Author Center, click on the "Manuscripts with Decisions" queue and then clicking on the "view decision letter" link for this manuscript. You must scroll down to the very bottom of the letter to see the file(s), if any. This will open the file that the reviewer(s) or the Associate Editor included for you along with their review.

Reviewer: 1

Recommendation: Reject

Comments:

Strengths

S1. Unlike differential privacy baselines that suffer severe accuracy degradation (DP-SGD collapses to <33% at $\epsilon \leq 1$), HE-IFD achieves 79.2% on CIFAR-10 (N=1) with cryptographic guarantees. The MIA AUC ≈ 0.5 empirically validates that the server gains zero information about client data — a meaningful and non-trivial contribution over prior plaintext federated distillation methods.

S2. The single-round encrypted upload reduces total communication cost from ~34 TB (multi-round encrypted FL at N=32) to ~460 GB, while keeping server-side computation constant at ~6.1 CPU-hours regardless of N. This decoupling of server computation from client count is practically attractive for large cross-silo deployments.

Weaknesses

W1. Accuracy drops sharply from 79.2% (N=1) to 35–37% (N=16) on CIFAR-10, tracking the Plaintext KL baseline. The paper attributes this to data fragmentation under Dirichlet heterogeneity but provides no mechanism to mitigate it, severely limiting practical utility in the large-N regime.

W2. The method requires replacing standard operators (ReLU $\rightarrow$ polynomial activations, BatchNorm $\rightarrow$ ChannelScale, MaxPool $\rightarrow$ identity), which imposes accuracy loss. The paper does not isolate and quantify this cost separately, making it difficult to assess the true contribution of the HE protocol itself.

W3. ~460 GB total upload per round is still prohibitive for real cross-silo deployments. The paper neither discusses bandwidth constraints in practice nor proposes compression strategies, leaving a significant engineering gap.

W4. Privacy arguments rely primarily on empirical attack simulations (LiRA, loss-threshold MIA) rather than formal end-to-end privacy proofs. Critically, the paper acknowledges that the released student model post-decryption may still leak client-specific information — a residual risk that is not adequately addressed. Pruning? discuss with Kerem

W5. Experiments are restricted to CIFAR-10 and FashionMNIST with small/medium models; no large-scale or NLP benchmarks. Key ablations are absent: individual contributions of magnitude regularization vs. affine bridges, sensitivity to the 10% feature extraction budget, and impact of threshold decryption on accuracy.

Additional Questions:

1. Which category describes this manuscript?: Research/Technology

2. How relevant is this manuscript to the readers of this periodical? Please explain your rating under Public Comments below.: Interesting - but not very relevant

1. Please explain how this manuscript advances this field of research and/or contributes something new to the literature.: This paper proposes HE-IFD, a one-shot federated learning framework that combines intermediate feature distillation with homomorphic encryption (CKKS scheme) to enable privacy-

preserving knowledge transfer from distributed client teacher models to a central server student model.

2. Is the manuscript technically sound? Please explain your answer under Public Comments below.: Partially

1. Are the title, abstract, and keywords appropriate? Please explain under Public Comments below.: Yes

2. Does the manuscript contain sufficient and appropriate references? Please explain under Public Comments below.: References are sufficient and appropriate

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.

If you are not suggesting any references, please type NA.: NA.

3. Does the introduction state the objectives of the manuscript in terms that encourage the reader to read on? Please explain your answer under Public Comments below.: Could be improved

4. How would you rate the organization of the manuscript? Is it focused? Is the length appropriate for the topic? Please explain under Public Comments below.: Could be improved

5. Please rate the readability of the manuscript. Explain your rating under Public Comments below.: Difficult to read and understand

6. Should the supplemental material be included? (Click on the Supplementary Files icon to view files): Does not apply, no supplementary files included

7. If yes to 6, should it be accepted: After revisions. Please include explanation under Public Comments below.

Please rate the manuscript. Explain your choice: Fair

Reviewer: 2

Recommendation: Author Should Prepare A Major Revision For A Second Review

Comments:

[Summary]

This paper proposes HE-IFD, a one-shot federated knowledge distillation framework in which clients encrypt intermediate feature pairs from locally trained teachers under CKKS and upload them once to a server. The server trains a HE-compatible student entirely over ciphertexts using a progressive block-by-block procedure. The authors report competitive accuracy on CIFAR-10 and FashionMNIST across multiple architectures and client heterogeneity levels, analyze communication/compute costs.

[Strengths]

+The system design and workflow are presented clearly.

[Weaknesses]

-Lacking sufficient comparison with related work.
-Critical experimental details are missing.

[Comments]

While this paper addresses an impactful problem at the intersection of FL privacy, one-shot communication, and practical FHE constraints, the current version of this paper requires substantial additional experimentation and evidence (see below) to meet the bar for TDSC. I recommend a major revision.

1) The most significant and most critical weakness of the current version is the absence of any direct comparison with previously published work. As a result, the improvements claimed by the paper cannot be assessed.

2) Did you execute any end-to-end training runs under CKKS with ciphertext weights and data (not just

forward passes or sub-operations)? If so, please report wall-clock times, rotation counts, memory usage, and convergence behavior for at least one block/small CNN.

3) Please quantify client-side encryption time and compute for the stated uploads (e.g., 115 GB per client at $N=4$). What is the per-sample encryption throughput and total time under your parameter settings?

4) What is the impact of malicious or colluding clients? Can a client poison the student via encrypted features? Do you envision robust aggregation or anomaly detection that is compatible with encryption?

5) In terms of post-release privacy, can you report MIA/inference attack results against the decrypted student as seen by a client or an external adversary, across N and α , to complement the “by construction” training privacy?

6) Why not keep student weights plaintext during server-side training (still protecting all client features/labels) to reduce CT×CT costs to CT×PT and avoid encrypted optimizer states? Would that still satisfy your threat model, given that weights are a function of pooled data?

Additional Questions:

1. Which category describes this manuscript?: Practice / Application / Case Study / Experience Report

2. How relevant is this manuscript to the readers of this periodical? Please explain your rating under Public Comments below.: Relevant

1. Please explain how this manuscript advances this field of research and/or contributes something new to the literature.: This paper propose a one-shot federated knowledge distillation framework in which all client contributions remain encrypted throughout server-side processing.

2. Is the manuscript technically sound? Please explain your answer under Public Comments below.: Partially

1. Are the title, abstract, and keywords appropriate? Please explain under Public Comments below.: Yes

2. Does the manuscript contain sufficient and appropriate references? Please explain under Public Comments below.: Important references are missing; more references are needed

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.

If you are not suggesting any references, please type NA.: NA

3. Does the introduction state the objectives of the manuscript in terms that encourage the reader to read on? Please explain your answer under Public Comments below.: No

4. How would you rate the organization of the manuscript? Is it focused? Is the length appropriate for the topic? Please explain under Public Comments below.: Could be improved

5. Please rate the readability of the manuscript. Explain your rating under Public Comments below.: Difficult to read and understand

6. Should the supplemental material be included? (Click on the Supplementary Files icon to view files): Does not apply, no supplementary files included

7. If yes to 6, should it be accepted: As is

Please rate the manuscript. Explain your choice: Fair

Reviewer: 3

Recommendation: Revise and resubmit as “new”

Comments:

Overall recommendation: Reject

This manuscript focuses on one-shot, privacy-preserving federated knowledge distillation framework under Homomorphic Encryption. It attempt to address the issue that training an HE-compatible student with polynomial activations and no batch normalization is unstable under standard end-to-end distillation. Combined with client-side collaborative normalization, magnitude regularized loss, and student-input training, this procedure yields stable convergence. However, despite the certain research significance of this work, there are several critical issues, especially in terms of writing, that need to be addressed. Therefore, I recommend it to be rejected. The problems are as follows:

1. The abstract mentions that "under non-IID data partitions, the student aggregates specialized knowledge from all teachers, matching or exceeding the performance that each teacher achieves on its own local class distribution. This presents a strong participation incentive for every client". However, the authors fail to provide sufficient justification or explanation for why this constitutes a strong incentive for clients. Add some more explanation
2. The challenges claimed in this paper and the contributions of this work are seriously mismatched. The authors do not clearly explain how each proposed contribution addresses the corresponding challenges raised in the manuscript. Write in a more direct manner, linking to chapters Firstly, in Section A-I, the authors present three challenges labeled polynomial magnitude explosion, training–distillation distribution gap, and scale-aligned distillation loss. Although partial solutions to challenges training–distillation distribution gap and scale-aligned distillation loss are introduced, no corresponding solution or countermeasure is provided for challenge polynomial magnitude explosion at all. Secondly, the paper lists four claimed contributions. Among them, the first describes the overall framework and the fourth introduce the empirical evaluation. The third seems to eliminates the challenge of polynomial magnitude explosion. However, the challenge corresponding to the second contribution is ambiguous. In addition, it is not clearly specified which contribution is used to address training–distillation distribution gap, and scale-aligned distillation loss respectively.
3. The manuscript presents the Motivation for One-Shot Communication and Homomorphic Encryption in the Section II-C. However, it should be discussed in the Introduction. Furthermore, it is unclear whether the subsequent discussions on HE and DP are related to the motivation or the related work, which makes readers confused.
4. The overview Fig.1 contains insufficient information and provides little help for understanding the proposed scheme.
5. In Section V-B, it only describes the results " HE-IFD consistently exceeds the mean teacher.". However, it does not explain how the proposed method works. Moreover, only the mean teacher model is compared, and no comparisons are made with other federated learning frameworks such as FedAvg, FedMD, etc., resulting in the lack of persuasiveness of the results.
6. In Section V-F it discusses the future directions. However, it does not fall into the scope of feasibility analysis.

Additional Questions:

1. Which category describes this manuscript?: Research/Technology

2. How relevant is this manuscript to the readers of this periodical? Please explain your rating under Public Comments below.: Relevant

1. Please explain how this manuscript advances this field of research and/or contributes something new to the literature.: This manuscript focuses on one-shot, privacy-preserving federated knowledge distillation framework under Homomorphic Encryption. It attempt to address the issue that training an HE-compatible student with polynomial activations and no batch normalization is unstable under standard end-to-end distillation. Combined with client-side collaborative normalization, magnitude regularized loss, and student-input training, this procedure yields stable convergence.

2. Is the manuscript technically sound? Please explain your answer under Public Comments below.: Appears to be - but didn't check completely

1. Are the title, abstract, and keywords appropriate? Please explain under Public Comments below.: Yes

2. Does the manuscript contain sufficient and appropriate references? Please explain under Public Comments below.: References are sufficient and appropriate

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.

If you are not suggesting any references, please type NA.: NA

3. Does the introduction state the objectives of the manuscript in terms that encourage the reader to read on? Please explain your answer under Public Comments below.: Could be improved

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Difficult to read and understand

6. Should the supplemental material be included? (Click on the Supplementary Files icon to view files):
Does not apply, no supplementary files included

7. If yes to 6, should it be accepted:

Please rate the manuscript. Explain your choice: Fair